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A Survey of Graphical Models for Decision-Making: Integrating Causality and Game Theory

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Abstract

Causality and game theory are two influential fields that contribute significantly to decision-making in various domains. Causality defines and models causal relationships in complex policy problems, while game theory provides insights into strategic interactions among stakeholders with competing interests. Integrating these frameworks has led to significant theoretical advancements with the potential to improve decision-making processes. However, practical applications of these developments remain underexplored. To support efforts toward implementation, this paper surveys key concepts in game theory and causality that are essential to their intersection, particularly within the context of probabilistic graphical models. By rigorously examining these concepts and illustrating them with intuitive, consistent examples, we clarify the required inputs for implementing these models, provide practitioners with insights into their application and selection across different scenarios, and reference existing research that supports their implementation. We hope this work encourages broader adoption of these models in real-world scenarios.

Keywords: Causal Modeling, Game Theory, Causal Game Theory, Probabilistic Graphical Models, Strategic Interaction, Decision-Making

1 Introduction

Causality involves defining and modeling causal relationships between variables, aiming to determine how changes in one variable influence another. The key distinction between causal and associative models is that causal models focus on intervention rather than mere observation (Spirtes et al, 2000; Gonzalez-Soto, 2021; Vonk et al, 2024). Once a causal relationship between two events is established, actively manipulating the cause enables more reliable outcome predictions compared to relying solely on correlations between observations. Developing a causal model of an environment and leveraging it for decision-making enables explanations beyond what an associative model can provide, as it allows one to ask why. Once learned, a causal model is independent of the agent and its preferences, making the acquired knowledge transferable to similar domains where understanding causal relationships proves beneficial (Pearl and Mackenzie, 2018; Gonzalez-Soto et al, 2023). Because of this practical significance and broad applicability, causal analysis has permeated diverse research fields, including medical treatment (Shalit, 2020), policy-making (Kreif and DiazOrdaz, 2019), social sciences (Sobel and Legare, 2014), epidemiology (Halloran and Struchiner, 1995), and cybersecurity (Andrew et al, 2022; Dhir et al, 2021).

On the other hand, game theory studies strategic interactions between rational decision-makers. It provides a framework to formalize the behavior resulting from interactions between competitive or cooperative agents. As a result, it has percolated the field of international security (Lindelauf et al, 2013), cyber security (Do et al, 2017), biology (Leimar and McNamara, 2023), and climate change (Wood, 2011).

The integration of causality and game theory builds upon decades of foundational work in probabilistic graphical models. The incipient step toward bringing these models into the decision-making domain was the development of influence diagrams (Miller III et al, 1976), later formalized by Howard and Matheson (2005). In parallel, influence diagrams were extended to multi-agent settings through multi-agent influence diagrams (MAIDs), providing a compact representation for games and enabling the computation of equilibrium strategies (Koller and Milch, 2003). Up to that point, however, these probabilistic graphical models did not incorporate explicit causal semantics. Since Koller and Milch’s approach was built upon Bayesian networks, and Bayesian networks had been independently extended to the causal setting by Pearl (2000), a natural synthesis emerged. Early work by Heckerman and Shachter (1994) had already proposed a decision-based view of causality within influence diagrams, laying the groundwork for causal influence diagrams (Dawid, 2002). More

recently, [Hammond et al \(2023\)](#) extended multi-agent influence diagrams to the causal realm, creating multi-agent causal influence diagrams (MACIDs) and developing formal machinery to reason about interventions in multi-agent systems. Concurrently, [Gonzalez-Soto et al \(2020\)](#) introduced agents' beliefs about causal structure into game-theoretic models, further enriching the integration.

The integration of causality and game theory offers distinct advantages that neither the game-theoretic nor the causal inference framework provides independently. Game-theoretic models capture strategic interdependencies and allow researchers to compute equilibrium strategies, identifying how rational agents best respond to one another's actions. However, these models typically remain silent on how the underlying environment can be manipulated and what would happen if the structure of the strategic interaction itself were altered. Causal models, conversely, formalize intervention, enabling researchers to ask not only what is associated with what, but what would happen if one were to actively intervene on a variable. Yet causal models traditionally assume a single decision-maker or treat other agents as part of the environment rather than as strategic actors with their own objectives.

By combining these perspectives, causal game theory enables researchers to reason about interventions in multi-agent settings where outcomes depend on others' strategic responses. Crucially, game theory alone cannot distinguish between observing an agent take an action and forcing that agent to take it. These yield different outcomes because forcing an action severs causal links that observation leaves intact. Causal game theory makes this distinction precise, allowing researchers to analyse how external interventions alter the strategic environment itself, both before and after equilibrium strategies are determined. This capacity to reason about changes to the game's structure, rather than merely computing responses within a fixed structure, is what the integration of causality with game theory uniquely provides.

Despite significant theoretical advances in causal game theory, practical applications remain limited. Several studies indicate that researchers often avoid employing causal and game-theoretic models due to difficulties in both comprehension and implementation ([Buseti, 2023](#); [Musci and Stuart, 2019](#); [Hermans et al, 2014](#)). This challenge is even more pronounced when integrating causal reasoning with game-theoretic modeling, as the absence of established frameworks complicates their combined application. Consequently, there have been increasing calls to bridge the gap between theory and practice through structured dialogues and closer collaborations between practitioners and methodologists ([Shubik, 2012](#); [Musci and Stuart, 2019](#)).

To address this gap, this paper provides a structured framework that synthesizes key concepts in causality, game theory, and their intersection within a unified probabilistic graphical model framework. By consistently adapting a practical example derived from security environments and referencing relevant research for implementation, we offer a concrete guide to navigating the complexities of integrating causal reasoning with game-theoretic modeling. This approach aims to bridge the divide between theory and practice, equipping

practitioners with structured guidance for selecting and applying these models and fostering closer collaboration between researchers and methodologists.

More specifically, we systematically map diverse game forms, including normal form, extensive form, and Bayesian games, to their corresponding causal representations, articulating the mathematical distinctions that clarify which causal and strategic structures are required for different analytical contexts. We focus on probabilistic graphical models (PGMs) as they provide an intuitive representation of dependencies (Pernkopf et al, 2014) and a capacity to unify diverse cognitive processes through a shared representational framework (Danks, 2014; Koller and Friedman, 2009; Sucar, 2020). Key concepts from causal game theory are elucidated, followed by an examination of the inputs required to operationalize such models. This detailed examination helps practitioners select the most suitable model for their specific application while clarifying the information required to work with these models effectively. We point to research that discusses techniques for eliciting such information and illustrate these concepts with examples involving the deterrence of adversarial attacks. Finally, we discuss further considerations and insights to help surmount practical implementation challenges. The conceptual scope of this paper

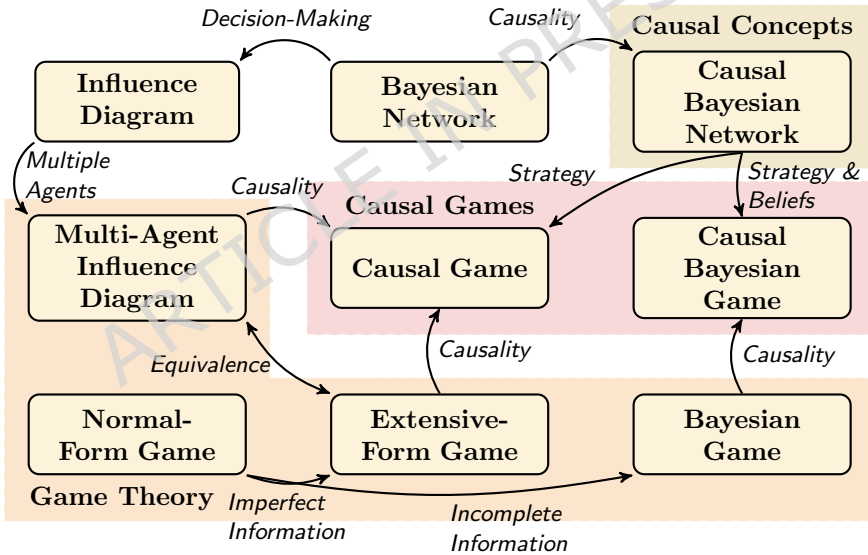


Fig. 1: Scope of this paper: the yellow blocks represent key concepts discussed within each domain: causality, game theory, and causal game theory. The concepts are grouped into categories that highlight their primary associations, indicated with background in different colors. Arrows indicate the possible extension or adaptation that allows for the transition from one concept to another.

is further illustrated in Figure 1, which categorizes the key concepts discussed across causality, game theory, and causal game theory.

Besides a practical guide to causal discovery (Malinsky and Danks, 2018a), there have been papers written about the construction of influence diagrams along with utilities (Bielza et al, 2010) and about representation restrictions imposed by influence diagrams (Bielza et al, 2009). To the best of our knowledge, our work is the first to address the causal game-theoretical models from a practical point of view.

Although doubts concerning the assumption of agent rationality (Sen, 1977; Gintis, 2000) have led to skepticism regarding the applicability of game-theoretic models (von Neumann and Morgenstern, 2004; Rubinstein, 2012; Rapoport and Orwant, 1962), our paper refrains from entering this philosophical debate.

The structure of this paper is as follows. Section 2 presents game-theoretic models, their solution concepts, and illustrative applications, along with a practical guide to their implementation. Section 3 explores causality, covering topics from causal discovery to causal inference and their practical applications. The intersection of causality and game theory is examined in Section 4. Finally, Section 5 reflects on the key insights and outlines directions for future research.

2 Game Theory

As the aim of this paper is to connect causality with game theory, we *restrict* ourselves to the game-theoretical components that have extensions in causality. Therefore, we reflect on three different types of games: the normal-form game, the extensive-form game, and the Bayesian game. A normal-form game models strategic interactions in which players select actions simultaneously without knowledge of other players' choices, making it well-suited for competitive scenarios such as pricing strategies. An extensive-form game represents sequential decision-making, where players act in a structured order, as in negotiations. A Bayesian game incorporates uncertainty, allowing players to make decisions based on private beliefs about unknown factors, making it particularly applicable to auctions.

After introducing preliminaries on game theory, we continue by outlining formal game definitions and relevant solution concepts. Furthermore, we explore the applicability of these game forms by analyzing similar examples across different scenarios and outline practical steps for modeling with these forms. Finally, we address the challenges associated with each form and discuss their practical utility. An overview of the game forms discussed, their concomitant characteristics, and required information for model implementation are summarized in Figure 4.

2.1 Preliminaries

A game models strategic interactions among rational decision-makers, called *agents* or *players*, where each decision-maker selects actions to maximize their own objectives while considering the choices of others. It provides a structured framework to analyze decision-making in competitive or cooperative settings.

Definition 1 (Game) A *game* is denoted by $\Gamma = (M, \mathbf{A}, \mathbf{U})$ and consists of:

- A finite set of *players* $M = \{1, \dots, m\}$.
- A set of *action sets* $\mathbf{A} = \{A^1, \dots, A^m\}$, where A^i is the set of available actions for player $i \in M$.
- A set of *utility functions* $\mathbf{U} = \{u^1, \dots, u^m\}$, where $u^i : \prod_{j \in M} A^j \rightarrow \mathbb{R}$ maps each action profile $(a^1, \dots, a^m)$ with $a^j \in A^j$ to a real-valued payoff for player i .

An *action profile* is a tuple $a = (a^1, \dots, a^m) \in \prod_{j \in M} A^j$ specifying one action for each player. For a given player i , the notation $a^{-i} = (a^1, \dots, a^{i-1}, a^{i+1}, \dots, a^m)$ denotes the *partial action profile* of all players except i , so that $a = (a^i, a^{-i})$.

Definition 2 (Strategy) A *strategy* (or *mixed strategy*) for player $i \in M$ is a probability distribution $\sigma^i \in \Delta(A^i)$, where $\Delta(A^i)$ denotes the simplex over the action set A^i . That is, σ^i assigns a probability $\sigma^i(a^i) \geq 0$ to each action $a^i \in A^i$ such that $\sum_{a^i \in A^i} \sigma^i(a^i) = 1$. The set of all strategies for player i is denoted by $\Sigma^i = \Delta(A^i)$. A strategy is called *pure* if $\sigma^i(a^i) \in \{0, 1\}$ for all $a^i \in A^i$, meaning the player deterministically selects a single action, and *fully mixed* if $\sigma^i(a^i) > 0$ for all $a^i \in A^i$, meaning every action is chosen with positive probability.

A *strategy profile* $\sigma = (\sigma^1, \dots, \sigma^m)$ specifies a strategy for each player. The expected utility for player i under a strategy profile σ is denoted $u^i(\sigma)$.

Definition 3 (Strategy Profile) A *strategy profile* is a tuple $\sigma = (\sigma^1, \dots, \sigma^m) \in \prod_{j \in M} \Sigma^j$ specifying one strategy for each player. For a given player i , the notation $\sigma^{-i} = (\sigma^1, \dots, \sigma^{i-1}, \sigma^{i+1}, \dots, \sigma^m)$ denotes the *partial strategy profile* of all players except i , so that $\sigma = (\sigma^i, \sigma^{-i})$.

2.2 Normal-Form Game

First, we write down the definition of a normal-form game and its associated solution concept. Then we provide an example.

Definition 4 A *normal-form game* (NFG) is a tuple $\Gamma = (M, \mathbf{A}, \mathbf{U})$ for which:

- $M = \{1, \dots, m\}$ is a set of agents.

- $\mathbf{A} = \{A^1, \dots, A^m\}$ is the set of action set, where A^i denotes the set of actions available to agent $i \in M$.
- $\mathbf{U} = \{u^1, \dots, u^m\}$ is a set of utility functions where $u^i : \mathbf{A} \rightarrow \mathbb{R}$ is the payoff function for agent $i \in M$, representing the payoff that agent i receives.

Definition 5 A strategy profile $\hat{\sigma} = (\hat{\sigma}^1, \dots, \hat{\sigma}^m)$ is a *Nash equilibrium* if for every player $i \in \{1, \dots, m\}$:

$$\hat{\sigma}^i \in \arg \max_{\sigma^i \in \Sigma^i} u^i(\sigma^i, \hat{\sigma}^{-i}).$$

where Σ^i is player i 's strategy space.

Example 1 (Deterring Game in Normal-Form) Suppose a game models a strategic interaction between a deterring agent and its attacking adversary. The deterring agent threatens retaliation if attacked, but whether the agent is willing to follow through (d) or is bluffing ($\neg d$) is determined by action A^1 . Simultaneously, the adversary decides whether to attack (a) or not ($\neg a$), denoted by action A^2 . The game can be illustrated by Figure 2a and the game matrix is displayed in Table 1. The two pure Nash equilibria are $(\neg a, d)$ and $(a, \neg d)$. These are identified by confirming that, in each case, neither player can improve their payoff by unilaterally switching strategies.

Several open-source solvers are available for computing Nash equilibria in normal-form games, including the Nashpy (Knight and Campbell, 2018) and Gambit (Savani and Turocy, 2024) libraries.

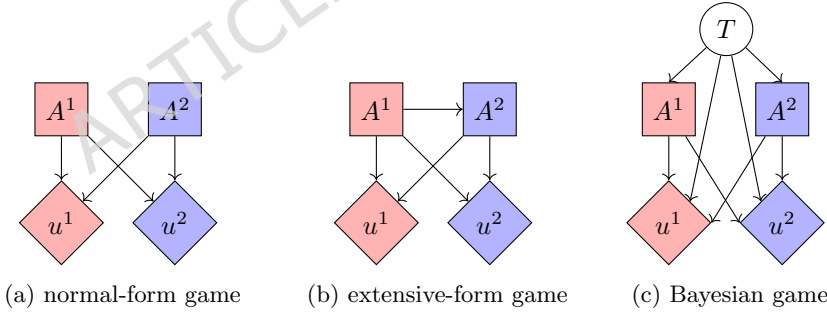


Fig. 2: The relations of the normal-form game (a), extensive-form game (b), and Bayesian game (c). Arrows represent information flow between decision nodes. These relations correspond to Example 1 for (a), Examples 2 and 3 for (b), and Example 4 for (c). In the normal-form game, the deterring and attacking agents make independent decisions that determine their utilities. In contrast, the attacker's decisions are shaped by the deterrer's actions in the extensive-form game. Lastly, in the Bayesian game, the agents' decisions and utilities are influenced by their individual types and their beliefs about the opponent's type.

Although the deterring agent acts first by issuing a threat of retaliation, the game-theoretical model does not consider the act of threatening as a strategic decision. Instead, the decision of interest is whether the deterring agent follows through on the threat. Since this decision does not occur after the attacker's move, both actions are effectively simultaneous and can be represented using a normal-form game. Nonetheless, it may very well be that acts do happen sequentially. In this case, a more refined game form is necessary, which is the extensive-form game.

2.3 Extensive-Form Game

First, we write down the definition of an extensive-form game, followed by an example and the relevance of a solution concept known as a subgame perfect equilibrium. We follow the definition of [Hammond et al \(2023\)](#).

Definition 6 An *extensive-form game* (EFG) is a tuple $\Gamma = (M, G, \mathbf{P}, \mathbf{A}, \lambda, \mathbf{I}, \mathbf{U})$ for which:

- $M = \{1, \dots, m\}$ is a set of agents.
- $G = (\mathbf{V}, \mathbf{E})$ is a rooted tree, where the nodes $\mathbf{V}$ are partitioned into sets $\mathbf{V}^0, \mathbf{V}^1, \dots, \mathbf{V}^n, \mathbf{T}$. In this case, $\mathbf{T}$ are the leaves or terminal nodes of G , $\mathbf{V}^0$ are chance nodes, and $\mathbf{V}^i$ are the decision nodes controlled by agent $i \in M$. The nodes are connected by edges $\mathbf{E}$.
- $\mathbf{P} = \{P_1, \dots, P_{|\mathbf{V}^0|}\}$ represents a set of probability distributions P_j defined over the children of each chance node V_j^0 , denoted as $\mathbf{ch}(V_j^0)$, for $j = 1, 2, \dots, |\mathbf{V}^0|$.
- $\mathbf{A}$ represents the set of action sets, where $A_j^i \subseteq \mathbf{A}$ indicates the set of actions available at $V_j^i \in \mathbf{V}^i$.
- $\lambda : \mathbf{E} \rightarrow \mathbf{A}$ is a labelling function that assigns each edge (V_j^i, V_l^k) to an action $a \in A_j^i$.
- $\mathbf{I} = \{\mathbf{I}^1, \dots, \mathbf{I}^m\}$ represents a collection of information sets, which partition the decision nodes controlled by agent i . Each information set $I_j^i \in \mathbf{I}^i$ is defined such that, for all $V_k^i, V_l^i \in I_j^i$, the available actions at these nodes are identical, i.e., $A_k^i = A_l^i$.
- $\mathbf{U} = \{u^1, \dots, u^m\}$ is a set of utility functions where $u^i : \mathbf{T} \rightarrow \mathbb{R}$ is the payoff function for agent $i \in M$, representing the payoff that agent i receives.

Table 1: Utilities in Deterring Game (or Game of Chicken) in Normal-Form.

		Adversary	
		a	$\neg a$
Deterrer	d	$(-1000, -1000)$	$(1, -1)$
	$\neg d$	$(-1, 1)$	$(0, 0)$

Example 2 (Deterring Game in Extensive-Form) Suppose a game models a strategic interaction between a deterring agent and its attacking adversary. This time, the deterring agent aims to deter by threatening retaliation denoted by action A^1 in node V_1^1 , which it is willing to follow through (d) or is bluffing ($\neg d$). Subsequently, the adversary then acts A^2 to decide whether to attack (a) or not ($\neg a$) in nodes (V_1^2, V_2^2). Since the adversary does not have knowledge about the credibility of the deterrence effort, both nodes (V_1^2, V_2^2) are in the same information set, I_1^2 . The dependencies of the game can be illustrated by Figure 2b and the game tree of Figure 3.

Should the adversary have full information about the action of the deterring agent, a new game arises in which the adversary evaluates its actions based on this knowledge. Formally, this is known as a *subgame*.

Definition 7 A *subgame* of an EFG is a game with a game tree $G' = (\mathbf{V}', \mathbf{E}')$ that is restricted to a node and its descendants such that any information set is either completely in the subgame or completely out the subgame.

Definition 8 A *subgame perfect equilibrium* is a strategy profile σ such that for every subgame, σ constitutes a Nash equilibrium of the subgame.

Subgame perfect equilibria are relevant for the exclusion of non-credible threats, which are threats that a rational player has no incentive to carry out in later stages of the game. The open-source solvers of Nashpy (Knight and Campbell, 2018) and Gambit (Savani and Turocy, 2024) compute the subgame perfect equilibrium using backward induction, starting at terminal nodes and working backward to ensure optimal play at each decision point. This eliminates Nash equilibria that rely on non-credible threats, which is illustrated by the following example:

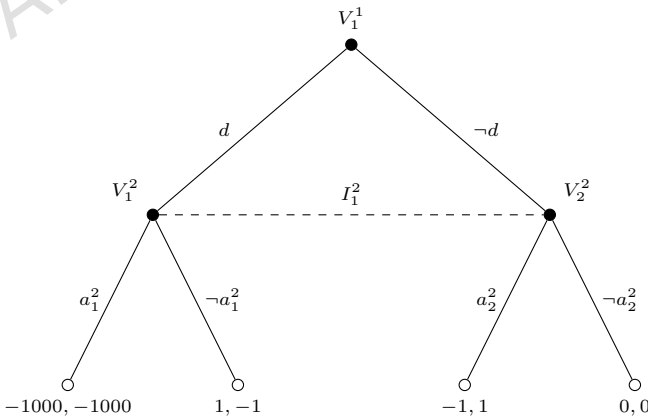


Fig. 3: Deterring Game in Extensive-Form

Example 3 (Deterring Subgame in Extensive-Form) Consider a slightly modified version of Example 2 under conditions of perfect information, where the adversary recognizes the deterrer's commitment to enforce the threat, as shown in Figure 2b. The Nash equilibria are $((-a_1^2, -a_2^2), d)$, $((-a_1^2, a_2^2), d)$, and $((a_1^2, a_2^2), -d)$.

To find the subgame perfect equilibrium via backward induction: First, at each of the adversary's decision nodes, determine the optimal response: after the deterrer chooses d , the adversary prefers $-a_1^2$ to avoid mutual destruction; after $-d$, the adversary prefers a_2^2 to exploit the lack of deterrence. Working backward to the root, the deterrer anticipates this behavior and chooses d , yielding payoff 1 rather than -1 from $-d$. This eliminates the Nash equilibrium $((a_1^2, a_2^2), -d)$, which relies on the non-credible threat that the adversary would attack even after deterrence is committed. The Nash equilibrium $((-a_1^2, -a_2^2), d)$ is also eliminated as it requires the adversary to irrationally choose $-a_2^2$ after $-d$. Thus, $((-a_1^2, a_2^2), d)$ is the unique subgame perfect equilibrium.

While *imperfect information* (uncertainty about the history of play) is captured directly through information sets, *incomplete information* (uncertainty about game structure) requires modeling player types and beliefs. Bayesian games provide the framework for this.

2.4 Bayesian Game

We begin with the formal definition of a Bayesian game, followed by the introduction of its solution concept, the Bayesian Nash equilibrium. Finally, we present an example that extends the previous examples.

Although sometimes introduced in terms of the nature states (Osborne and Rubinstein, 1994), we adopt the Bayesian game in terms of players' types as originally proposed (Harsanyi, 1967; Fudenberg and Tirole, 1991).

Definition 9 A *Bayesian game* is a tuple $\Gamma = \{M, \mathbf{A}, T, P, \mathbf{U}\}$, such that:

- $M = \{1, \dots, m\}$ is the finite player set.
- $\mathbf{A}$ represents the set of action sets, where A^i is the action set of player i for $i \in M$.
- T^i is the finite type set of player i , and $t^i \in T^i$ its type. $T = (T^1, \dots, T^m)$ is called the type profile tuple of Γ .
- $P : T \rightarrow [0, 1]$ is a probability distribution over T , referred to as the common prior. The belief of player i is denoted by

$$P(t^{-i} | t^i) = \frac{P(t^{-i}, t^i)}{P(t^i)} = \frac{P(t^{-i}, t^i)}{\sum_{t^{-i}} P(t^{-i}, t^i)},$$

which describes player i 's uncertainty about the other $m-1$ players' possible types t^{-i} , given player i 's type t^i , where $t^{-i} = (t^1, \dots, t^{i-1}, t^{i+1}, \dots, t^m)$ represents the tuple of the types of all the players except for player i .

- $U = \{u^1, \dots, u^m\}$ is a set of utility functions, where $u^i : T \times \mathbf{A} \rightarrow \mathbb{R}$ is the payoff function, which maps each action profile $\mathbf{a} \in \mathbf{A}$ to the pay-off of player i under each type profile $t^i \in T^i$.

Now that the structure of the game is contingent on the types of the players, the concept of a behavioral strategy can naturally be extended to account for these types: $\sigma^i(t^i) := \sigma^i(A^i \mid t^i)$ (Bajoori et al, 2016). This allows the definition of a *Bayesian Nash equilibrium*.

Definition 10 A strategy profile $\hat{\sigma} = (\hat{\sigma}^1, \dots, \hat{\sigma}^m)$ is a *Bayesian Nash equilibrium* if for every player $i \in M$ and type $t^i \in T^i$:

$$\hat{\sigma}^i(t^i) \in \arg \max_{\sigma^i \in \Sigma^i} P(t^{-i} \mid t^i) u^i(t^i, t^{-i}, \sigma^i, \hat{\sigma}^{-i}(t^{-i})).$$

Although the Harsanyi transformation allows Bayesian Nash equilibria to be computed by reformulating payoffs in terms of type-contingent strategies (Harsanyi (1967)), this approach scales exponentially with the number of types, motivating the development of approximate methods (Huang et al, 2025).

Example 4 (Deterring Game in Bayesian Form) Similar to the normal-form game, we model the actions of the deterring agent A^1 and its adversary A^2 independently. This time, the adversary can assume different types, where it is either protected against retaliation (t^2) or not (t^1). The relations of the game are illustrated by Figure 2c and the game matrices displayed in Table 2. The Bayesian Nash equilibria are $((\neg a, a), d)$ and $((a, a), \neg d)$. These are found by verifying that the deterrer maximizes expected utility given the probability distribution over adversary types ($p(t^1) = \frac{1}{4}$, $p(t^2) = \frac{3}{4}$), while each adversary type simultaneously maximizes their own payoff given their private information and the deterrer's strategy.

Table 2: Utilities in Deterring Game in Bayesian Game.

		Adversary Type $t^1: p = \frac{1}{4}$		Adversary Type $t^2: p = \frac{3}{4}$	
		a	$\neg a$	a	$\neg a$
Deterrer	d	$(-1000, -1000)$	$(1, -1)$	$(-1, 0)$	$(1, -1)$
	$\neg d$	$(-1, 1)$	$(0, 0)$	$(-1, 1)$	$(0, 0)$

2.5 Practical Guide of Game Theory

Game theory offers a structured framework for organizing information on actors' decision-making processes (Hermans et al, 2014). Before applying game-theoretic concepts, a qualitative process should define the specific rules of

the game for a given problem. This involves identifying stakeholders, outlining potential policy options, and establishing their interdependencies. Such an approach to framing policy problems is known as metagame analysis (Howard, 1971).

Practitioners should select the game type that best fits the policy problem's structure. Simultaneous decisions suit normal-form and Bayesian games,¹ whereas sequential decisions align with extensive-form games. Additionally, extensive-form games may involve imperfect information, while Bayesian games feature incomplete information. Although selecting a specific game type is useful, some scholars argue that a robust analytical approach benefits from modeling pluralism rather than strict uniformity (Allan and Dupont, 1999), advocating for the use of diverse game-theoretic frameworks where necessary while ensuring that variation remains focused on essential structural distinctions.

As the mathematical dissection of different games shows, each game requires specific information to be gathered before implementation. All games need the elicitation of utilities or preferences, with extensive-form and Bayesian games requiring more detailed utility structures. While utility elicitation is a demanding task, it has been well-studied and we refer the reader to the work

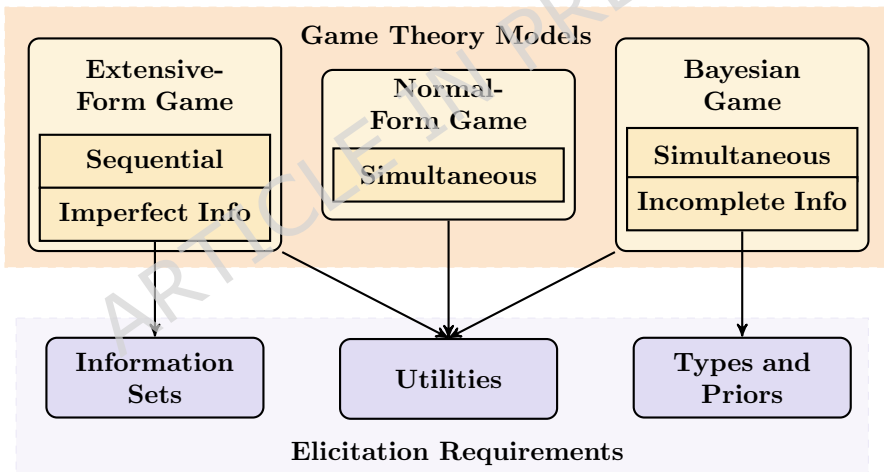


Fig. 4: Elicitation requirements for different game theory models: while the yellow blocks correspond to the different games along with their associated properties, the purple blocks indicate the different pieces of information that are required to be elicited. The arrows indicate what information is relevant to each game.

¹Bayesian games can also be extended to sequential decision-making (Osborne and Rubinstein, 1994).

by Wakker and Deneffe (1996) for details about elicitation methods. If preference relations are uncertain, alternative methods can still provide insights into the stability of certain solution concepts (Li et al, 2004).

In extensive-form games, information sets, clarifying who knows what at each decision point, need to be extracted. Additionally, chance nodes require probability distributions for the uncertainties they introduce. For Bayesian games, prior and posterior probabilities of types must also be elicited, a challenging process with guidelines written by Mikkola et al (2023). The characteristics of the different games, along with the required elicitation information can be observed in Figure 4.

To bridge the gap between theory and application in game theory, a growing body of work focuses on deriving game-theoretic models from simulation data to enhance empirical validity. A comprehensive survey of recent advances in this area is provided by Wellman et al (2024).

3 Causality

The concept of causality pertains to the underlying regularities within a given environment (or context) that go beyond mere probabilistic (or associative) relationships. A causal relationship enables assessing how a change in the cause affects the outcome, thereby establishing a directional order between causally linked variables (Spirtes et al, 2000; Gonzalez-Soto, 2021). Attempts to understand the meaning of cause have troubled philosophers and scientists, from David Hume and John Stuart Mill to modern scientists as Suppes (Suppes, 1973), Cartwright (Cartwright, 1983), Spirtes (Spirtes et al, 2000), Pearl (Pearl, 2009b), Spohn (Spohn, 2012). We adopt the manipulationist interpretation of causality (Spirtes et al, 2000; Woodward, 2005), as this framework is also adopted in studies of the intersection between causality and game theory (Gonzalez-Soto et al, 2020; Hammond et al, 2023). The main paradigm of this interpretation has been expressed by Cook et al as: manipulation of a cause will result in a corresponding change in the effect. Similar descriptions of the manipulationist approach can be found in the work of Holland. Bayesian networks provide the dominant graphical framework for representing such causal relationships. Beyond Bayesian networks, chain event graphs (CEGs) offer richer representations for asymmetric state spaces (Smith and Anderson, 2008), with extensions to causal analysis (Thwaites et al, 2010) and Bayesian games (Thwaites and Smith, 2018), though their integration with game-theoretic causality remains nascent (Ramiah et al, 2025).

We *restrict* ourselves to the aspects of causality that are most relevant to the specification and practical utility of such models.² Therefore, we first discuss some essential preliminaries. We continue with essential concepts such as the causal Bayesian network, the interventional distribution, and its associated truncated factorization in Section 3.2. These concepts help to formalize

²Although some game-theoretical advances have been made at the counterfactual layer of Pearl's Causal Hierarchy (Hammond et al, 2023; Bareinboim et al, 2022), we exclude them here as their practical utility is under scrutiny (Bareinboim et al, 2022).

causal structures, define causal interventions, and compute the effects of interventions, respectively. We then elaborate on the process of learning causal graphs, known as causal discovery, in Section 3.3 and continue with inferring causal effects, known as causal inference, in Section 3.4. The focus of the latter two sections will be on their practical considerations, which are summarized in Figure 5.

3.1 Preliminaries

A graph is denoted by $G = (\mathbf{V}, \mathbf{E})$, where $\mathbf{V} = \{V_1, \dots, V_n\}$ is the set of vertices (or nodes) and $\mathbf{E}$ the set of edges. A graph can be *directed* when every edge has a direction, *undirected* when no edge has a direction or *partially directed* when some but not all edges have a direction. A graph can contain a *cycle* when there exists a directed path from a node to itself. When there is no such path and the graph is directed, we call this a *directed acyclic graph* (DAG). Edges can also be *bidirected*. A graph containing only directed and bidirected arrows without directed cycles is called an *acyclic directed mixed graph* (ADMG). When there is a directed edge from node V_1 to node V_2 (or $V_1 \rightarrow V_2$, in short), we say that V_1 is a *parent* of V_2 and V_2 a *child* of V_1 . The set of parents of V_2 is denoted by $\mathbf{pa}(V_2)$ and the set of children of V_1 is denoted by $\mathbf{ch}(V_1)$.

When graphs are endowed with probabilistic meaning, random variables $\mathbf{X} = \{X_1, \dots, X_n\}$ will correspond to nodes of the graph $\mathbf{V} = \{V_1, \dots, V_n\}$ and therefore $\mathbf{V}$ will inherit the probability distributions and state spaces from $\mathbf{X}$ (meaning $P(\mathbf{V})$ and v_i will correspond to $P(\mathbf{X})$ and x_i , respectively). In this case, $\mathbf{pa}(V_i)$ refers to the random variables that are associated with the parents of V_i . The assignment of random variables $\mathbf{pa}(V_i)$ is denoted by $\mathbf{pa}_i$.

3.2 Bayesian Network and Interventional Distribution

A *Bayesian network* (BN) represents a distribution over a set of random variables by associating it to a graph in such a way that each of the random variables corresponds to a node in the graph, and the edges describe the probabilistic dependencies found in the distribution. Formally, a Bayesian network contains a directed acyclic graph G whose nodes represent random variables $X_1, \dots, X_n$ and a probability distribution P that factorizes over G , which means that the joint probability distribution of random variables $X_i \in \mathbf{X}$ can be expressed as

$$P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i \mid \mathbf{pa}_i).$$

This factorization of the distribution implies that every random variable X_i is conditionally independent of its non-descendants given its parents in the graph.

A *causal Bayesian network* is a BN whose edges represent relations of cause-effect. Also, the model is enriched with an operator named $do()$, which is defined over graphs, and whose action is described as follows.

Definition 11 Let $\mathbf{X}_S$ and $\mathbf{X}_K$ be sets of random variables. The *interventional distribution* $P(\mathbf{x}_S \mid do(\mathbf{X}_K = \mathbf{x}_K))$ encodes the probability that $\mathbf{X}_S = \mathbf{x}_S$ given that $\mathbf{X}_K$ is forced to take value $\mathbf{x}_K$ (denoted by the *do-operator* $do(\mathbf{X}_K = \mathbf{x}_K)$, or $do(\mathbf{x}_K)$ in short) with probability 1.

The behavior of the *do-operator* within a causal BN leads to a *truncated factorization* of the distribution (Bareinboim et al, 2012):

Definition 12 Let $P(x_1, \dots, x_n)$ be the joint probability distribution of random variables $X_i \in \mathbf{X}$ corresponding to the nodes $V_i \in \mathbf{V}$ in the directed acyclic graph $G = (\mathbf{V}, \mathbf{E})$. Suppose $\mathbf{X}_K, \mathbf{X}_S \subset \mathbf{X}$ and $\mathbf{X}_K \cap \mathbf{X}_S = \emptyset$. Then:

$$P(\mathbf{x}_S \mid do(\mathbf{X}_K = \mathbf{x}_K)) = \sum_{x_j \mid j \notin \mathbf{K} \cup \mathbf{S}} \prod_{j \notin \mathbf{K}} P(x_j \mid \mathbf{pa}_j).$$

Although the introduced definitions pertain to *Markovian* Causal Bayesian networks, they also extend naturally to scenarios involving *semi-Markovian* models where unobserved confounding variables are present. In this case, confounding relations can be represented by bidirected arrows in an ADMG and a similar truncated factorization can be derived (Richardson, 2014). Next, we demonstrate the application of the truncated factorization by refining our previous example within a causal framework.

Example 5 (Deterring Relations in Causal Form) Suppose we have observational data on the explicitness of deterrence messages X_D , which can either be explicit (d) or vague ($\neg d$). The goal of an explicit message is to dissuade the adversary from committing to an aggressive operation X_A ($X_D \rightarrow X_A$). However, both the explicitness of the deterrence message and the adversary's decision to attack are shaped by the deterring agent's military and strategic capabilities X_C , which can be strong (c) or weak ($\neg c$). These capabilities influence the explicitness of the message $X_C \rightarrow X_D$, but also directly affect the adversary's decision to commit to an aggressive operation $X_C \rightarrow X_A$. The causal relations are displayed in Figure 6a.

To compute the causal effect of explicit deterrence messaging on successful dissuasion, the covariate, the deterrer's capabilities, should be adjusted for. Therefore, the *do-operator* can be deployed, and the truncated factorization can be utilized:

$$P(X_A \mid do(X_D = d)) = \sum_{x_C \in \{c, \neg c\}} P(X_A \mid X_D = d, X_C = x_C) P(X_C = x_C).$$

3.3 Practical Guide to Causal Discovery

The goal of causal discovery is to recover the causal structure $G = (\mathbf{V}, \mathbf{E})$, either from observational data, interventional data, or a mixture of both. While there already exists a plethora of survey papers on causal discovery (Zanga et al, 2022; Nogueira et al, 2022; Assaad et al, 2022), including an exhaustive list of assumptions (Vonk et al, 2023) and a practical guide to causal discovery (Malinsky and Danks, 2018b), we would like to limit the discussions to the following *three* practical points when bringing causal discovery algorithms into practice.

First, when engaging in causal discovery, practitioners should abide by different assumptions. Causal sufficiency, the assumption that the dataset in question contains all common causes of the variables included, bifurcates different methods of causal discovery. Sometimes, assumptions can be replaced by others. For example, faithfulness, the assumption that dependencies and independencies in the data are solely due to the underlying causal structure, is considered a strong assumption (Andersen, 2013) and can be replaced by assumptions on the data-generating process of the graph (Shimizu et al, 2006). Therefore, it is recommended to check the validity of the assumptions per application and choose the appropriate causal discovery algorithm accordingly.

Second, the quality and the size of the available data should be leading when bringing the causal discovery methods into practice. Causal discovery algorithms are mostly grouped into constraint-based methods or score-based methods. While constraint-based methods exploit the conditional independencies to inform the structure of the graph (Spirtes et al, 1990; Spirtes and Glymour, 1991), score-based methods iteratively adjust the graph, scoring and comparing different structures to find the best fit to the data (Chickering, 2003). As both methods rely on statistical tests, larger sample sizes are typically preferred, because they provide more reliable estimates and increase the power of the tests. Nonetheless, score-based methods are preferred in the case of small sample sizes (Malinsky and Danks, 2018b). Despite the goal of causal discovery to specify the dependency *between* the random variables, the observations in the available sample are assumed to be independently distributed in order for these causal discovery algorithms to work (Vonk et al, 2023).

Finally, research has shown that causal discovery algorithms can be unstable (Kitson and Constantinou, 2024) or that only limited parts of the causal graph can be discovered from pure observations (Peters et al, 2017). To this purpose, domain knowledge can be used to refine the performance of causal discovery algorithms and has been incorporated via tiered background knowledge (Andrews et al, 2020) user interactions (Mäkelä et al, 2022) or the penalization of the search process (Hasan and Gani, 2022). It is recommended that practitioners assess the possibility of incorporating domain knowledge or experts to enhance the quality of the obtained causal graphs.

3.4 Practical Guide to Causal Inference

Causal inference is concerned with the estimation of an outcome variable under possible interventions. Comprehensive theoretical treatments exist elsewhere (Ding and Li, 2018; Yao et al, 2021; Pearl, 2009a), and we do not attempt to replicate them here. Instead, consistent with this paper’s aim of bridging theory and practice, we highlight practical challenges that arise when applying well-established theoretical frameworks to real-world data. We first discuss forms of bias that arise when identification assumptions are violated, and then turn to forms of bias that arise when implementing causal inference workflows.

Valid causal inference from observational data requires *identification*, the process of expressing a causal quantity in terms of observable probability distributions. This process rests on several assumptions whose violations introduce systematic bias. *Unmeasured confounding* occurs when variables that jointly influence both treatment and outcome are not observed or adjusted for, leaving open backdoor paths that conflate causal effects with spurious associations (Rosenbaum, 2002; VanderWeele and Ding, 2017). *Collider bias* arises when conditioning on a variable influenced by both treatment and outcome, opening non-causal pathways that can create or mask associations (Greenland, 2003; Elwert and Winship, 2014). Unlike random error, these biases persist regardless of sample size and can only be addressed by specifying a correct causal structure and conducting sensitivity analyses to assess robustness to potential misspecification.

While a correct causal structure is essential for valid identification, obtaining this structure introduces its own challenges. When causal discovery methods are used to learn the graphical structure from data, and causal inference is subsequently applied to the same data, *double-dipping bias* compromises the validity of confidence intervals and hypothesis tests (Cheng et al, 2024). This issue is analogous to model selection bias in traditional statistics: uncertainty estimates assume a fixed, pre-specified model, not one learned from the data. Practitioners should, where possible, apply correction methods (Gradu et al, 2024) or reserve a held-out dataset for inference.

Second, when data are unavailable for estimating conditional probability distributions, or when avoiding double dipping is paramount, practitioners may elicit these distributions from domain experts. While efficient methods exist for discrete distributions (Barons et al, 2022; Alkhairy et al, 2020; Hassall et al, 2019), elicitation of continuous distributions remains underdeveloped. Moreover, expert elicitation is susceptible to cognitive biases such as overconfidence and anchoring (Tversky and Kahneman, 1974; O’Hagan, 2019), introducing *elicitation bias*. Sensitivity analyses can help assess how robust conclusions are to plausible variations in elicited quantities.

Finally, the *positivity assumption* requires that every subgroup defined by confounders has a non-zero probability of receiving each treatment level. In practice, adjusting for the covariates required by the causal structure can substantially reduce effective sample size or result in positivity violations, where certain covariate combinations have no observed outcomes under one

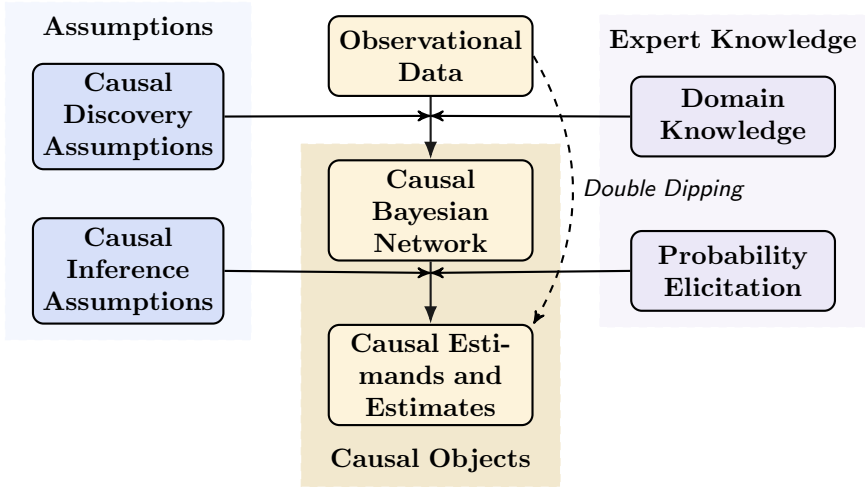


Fig. 5: Flow of causal reasoning from data to graphical components (causal discovery) and subsequently to causal estimands and estimates (causal inference). While the light blue part indicates at what stage assumptions have to be taken into account, the purple part indicates possible supplementary information from domain experts. Double dipping occurs when data in the causal discovery stage is being reused at the causal inference stage.

or more treatment conditions (Petersen et al, 2012; Westreich and Cole, 2010; D’Amour et al, 2021). Non-parametric methods are particularly sensitive to these issues, while parametric approaches require potentially restrictive distributional assumptions. Semi-parametric methods offer a middle ground, providing robustness while maintaining reasonable efficiency (Bhattacharya et al, 2022; Luo et al, 2023).

4 Game Theory and Causality

In this section, we explore the intersection of game theory and causality by integrating causal concepts from the previous section into the decision-making framework. Specifically, we extend Bayesian networks to influence diagrams (Howard and Matheson, 2005), distinguishing between purely probabilistic structures and decision-theoretic elements. This framework is further generalized to multi-agent settings, leading to multi-agent influence diagrams (Koller and Milch, 2003; Hammond et al, 2023), which form the foundational structure of causal games. To account for uncertainty in the causal structure of a game, we adapt the approach of Gonzalez-Soto et al (2025), introducing the causal Bayesian game. Finally, we discuss key considerations for the practical implementation of these models.

4.1 Influence Diagram

An influence diagram extends a Bayesian network to the decision realm by dissecting the nodes into chance nodes, utility nodes, and decision nodes. More formally:

Definition 13 An *influence diagram* contains a graphical structure $G = (\mathbf{V}, \mathbf{E})$ where $\mathbf{V}$ is separated into decision nodes $\mathbf{D}$, chance nodes $\mathbf{X}$ and utility nodes $\mathbf{U}$. Whereas the conditional probability distributions of $\mathbf{X}$ and $\mathbf{U}$ are known, any *decision rule* $\sigma(D)$ with $D \in \mathbf{D}$ corresponds to a conditional probability distribution over the decisions and hence all decision rules constitute the full joint probability distribution $P(\mathbf{V})$.

Example 6 (Deterring Relations as Influence Diagram) A refinement of Example 5 involves separating the chance node of the military and strategic capabilities X_C from the decision node of deterrence messaging D_C . In this framework, the aggressor's decision to conduct an aggressive operation X_A can be modeled as another chance node, which is followed by a final utility node of the deterring agent U^1 . These relations are illustrated by Figure 6b.

Influence diagrams incorporate decision-making elements into Bayesian networks,³ but they only model the decision-making of a single agent, meaning there is no strategic interaction between agents. Strategic considerations emerge only when multiple agents make decisions in response to each other, as seen in multi-agent influence diagrams.

4.2 Multi-Agent Influence Diagram And Causal Game

We start by introducing multi-agent influence diagrams (Koller and Milch, 2003) and continue with the definition of a causal game (Hammond et al, 2023). The latter concept will then be illustrated with an example.

Definition 14 A *multi-agent influence diagram* (MAID) contains a graphical structure $G = (\mathbf{V}, \mathbf{E})$ and a set of agents $M = \{1, \dots, m\}$. Furthermore, the nodes $\mathbf{V}$ are separated in decision nodes $\mathbf{D} = \cup_{i \in M} \mathbf{D}^i$, chance nodes $\mathbf{X}$ and utility nodes $\mathbf{U} = \cup_{i \in M} \mathbf{U}^i$. Each strategy $\sigma^i(D^i)$ with $D^i \in \mathbf{D}^i$ defines a conditional probability distribution over a decision node. Consequently, given conditional probability distributions of $\mathbf{X}$ and $\mathbf{U}$, a complete strategy profile σ constitutes the full joint probability distribution $P(\mathbf{V})$.⁴

The causal game Γ associated with a MAID can be seen as a more abstract form of a MAID, where the parameters of the decision variables are yet to be defined.

³In a similar way, causal Bayesian networks can be extended to *causal influence diagrams* (Fox et al, 2021).

⁴Since EFGs and MAIDs are proven to be equivalent (Hammond et al, 2023), we use σ again for a strategy profile.

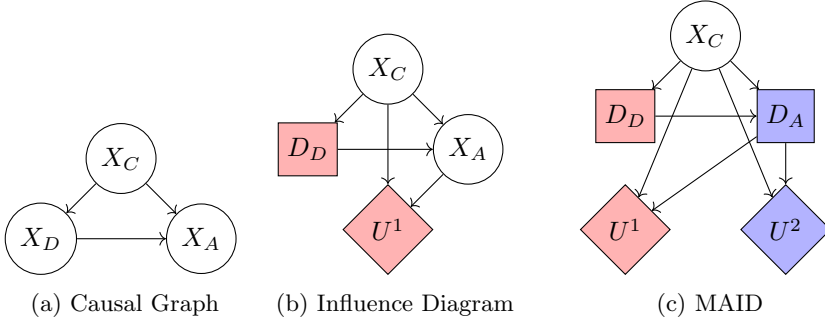


Fig. 6: The relations of the causal graph (a), influence diagram (b), and multi-agent influence diagram (c) of Example 5, 6 and 7, respectively. The causal graph does not distinguish between chance, decision and utility nodes as do the influence diagrams. In addition, the multi-agent influence diagram models the decision of the adversary also strategically. Arrows in causal graphs are inherently causal, while those in (multi-agent) influence diagrams may represent dependence or causation, with the latter requiring a formally defined do-operator (Hammond et al, 2023). We henceforth assume causal semantics for all influence diagrams.

Definition 15 A *causal game* Γ is a MAID such that for any chosen strategy profile σ , the induced joint probability distribution $P^\sigma(\mathbf{V})$ corresponds to a causal Bayesian network.

Similarly to extensive-form games, the Nash equilibrium of causal games can be defined in terms of the strategy profiles:

Definition 16 A strategy profile $\hat{\sigma} = (\hat{\sigma}^1, \dots, \hat{\sigma}^m)$ is a *Nash equilibrium* if for every player $i \in \{1, \dots, m\}$:

$$\hat{\sigma}^i \in \arg \max_{\sigma^i \in \Sigma^i} \sum_{U \in \mathbf{U}^i} \mathbb{E}_{[\sigma^i, \hat{\sigma}^{-i}]}[U].$$

Multi-agent influence diagrams are powerful models, because they allow for the calculation of equilibria as well as the computation of policy interventions.

Example 7 (Multi-Agent Influence Diagram Deterring Game) A multi-agent influence diagram can be derived from Example 6 when the adversary’s decision to conduct an aggressive operation a or refrain from one $\neg a$ is modeled as a decision node of another agent D_A . This decision node is influenced by the deterring agent’s decision node, D_D , which represents whether the deterrence messages are explicit (d) or vague ($\neg d$). Alongside the deterring agent’s utility node U^1 , the adversary also possesses a utility node U^2 . Both utility nodes depend on the attacker’s decision D_A and the deterrer’s capabilities X_C , which can be either strong (c) or weak ($\neg c$),

Fig. 7: The utilities of the deterring agent (left) and attacking agent (right).

U^1	$X_C = c$	$X_C = \neg c$	U^2	$X_C = c$	$X_C = \neg c$
$D_A = a$	0	-1	$D_A = a$	-1000	1
$D_A = \neg a$	1	1	$D_A = \neg a$	-1	-1

with an equal probability distribution. The relations are displayed in Figure 6c and the utilities are further specified in Figure 7.

This game admits eight Nash equilibria, identified by verifying that no agent can improve its expected utility through a unilateral deviation. The multiplicity arises because the defender’s payoff is independent of its messaging strategy: since changing the message alone affects neither agent’s utility, each of the two attacker best-response profiles extends across all four defender pure strategies. Four of these equilibria involve non-credible threats, where the attacker commits to attacking even against strong capabilities, sustained only because the defender’s strategy makes this contingency unreachable.

One equilibrium, $\hat{\sigma}$, prescribes that the deterring agent sends an explicit message when strong and a vague message otherwise, while the adversary attacks if and only if capabilities are weak. This equilibrium is also subgame perfect, obtained by backward induction: starting from the attacker’s decision, we determine its optimal response for each capability-message pair, then work backwards to the defender’s optimal messaging strategy, thereby eliminating all non-credible threats.⁵

Within this equilibrium, we can compute conditional expectations under the equilibrium strategy profile, such as the expected utility of the deterring agent for sending out an explicit message is $\mathbb{E}_{[\hat{\sigma}]}[U^1 \mid D_D = d] = 1$. We can also assess intervention effects in this equilibrium; for instance, if allied agents force an explicit deterrence message regardless of its capability, the utility becomes $\mathbb{E}_{[\hat{\sigma}]}[U^1 \mid do(D_D = d)] = 0$.

The PyCID package in Python (James Fox et al, 2021) provides tools for computing Nash equilibria, subgame perfect equilibria, and expected utilities under possible interventions in MAIDs.

This example represents a *post-policy* intervention, as results are computed after agents have settled into equilibrium behavior. In contrast, *pre-policy* interventions allow agents to adjust their strategies after an intervention is announced, computing a new equilibrium in the modified game. Both types of analysis are enabled by the causal structure of the model, which explicitly represents dependencies that interventions can modify. Note that arrows represent direct causal relationships, while indirect effects arising from game-theoretic reasoning (e.g., equilibrium adjustments) are not explicitly depicted but emerge from the model’s solution. Standard game-theoretic models, lacking this causal representation, cannot formally distinguish between pre and post-policy scenarios. A more refined treatment of pre-policy interventions can be found in Hammond et al (2023); a full introduction is beyond the

⁵Although the subgame perfect equilibrium of Definition 8 extends naturally to causal games, the notion of subgames in this setting is richer than in extensive-form games. See Hammond et al (2023) for details.

scope of this paper as they do not bring additional implications for practical implementation.

4.3 Causal Bayesian Games

The notion of a causal Bayesian game (Gonzalez-Soto et al, 2019, 2025) was developed to model settings where agents are uncertain about the causal structure governing their environment. This uncertainty belongs to the agents themselves: each agent may hold different beliefs about which causal model describes the true data-generating process, and since causal interventions produce different outcomes under different structures, expected payoffs become contingent on the underlying causal model. We refine the definition and notation of Gonzalez-Soto et al to align with the previously introduced causal games while ensuring consistency with earlier introduced Bayesian games. Following the Bayesian game in Section 2.4, the types correspond to distinct MAIDs within the family of causal graphical structures $\mathcal{G}$. Moreover, as in Bayesian games, players act independently, implying that we consider a subset of MAIDs where no direct causal paths exist between the decision nodes of different agents. Unlike the earlier introduced Bayesian game, players agree on the common state space $\mathcal{G}$ of possible graphical models but have private beliefs about the probability of these states $\mu_i(\mathcal{G})$. Naturally, interventions on decision nodes induce variations in payoffs across different graphical models $G \in \mathcal{G}$.

Definition 17 Consider a family of different causal structures $G \in \mathcal{G}$ where no direct paths exist between the decision nodes of different agents. Each agent $i \in \{1, \dots, m\}$ has a private belief about the probability of these causal structures $\mu_i(\mathcal{G})$ and a higher-order belief $\mu_i(\mu_{-i}(\mathcal{G}))$, which reflect uncertainty over other $m - 1$ players' beliefs. Each strategy $\sigma^i(G) = \sigma^i(D^i \mid G)$ defines a conditional probability distribution over a decision node $D^i \in \mathbf{D}^i$ conditional on the belief $\mu_i(G)$ of that graphical model $G \in \mathcal{G}$ for agent $i \in \{1, \dots, m\}$.⁶ A *causal Bayesian game* Γ is a MAID such that for any chosen strategy profile σ , the induced joint probability distribution $P^\sigma(\mathbf{V})$ corresponds to a causal Bayesian network.

Note that the causal Bayesian network is not only induced by the strategies for each player but by the strategies of the players conditioned on the same graphical model. Naturally, these considerations are also reflected in the Bayesian Nash equilibrium.

Definition 18 A strategy profile $\hat{\sigma} = (\hat{\sigma}^1, \dots, \hat{\sigma}^m)$ is a *Bayesian Nash equilibrium* if for every player $i \in \{1, \dots, m\}$ and graphical structure $G \in \mathcal{G}$:

$$\hat{\sigma}^i(G) \in \arg \max_{\sigma^i \in \Sigma^i} \mu_i(\mu_{-i}(G)) \sum_{U \in \mathbf{U}^i} \mathbb{E}_{[\sigma^i(G), \hat{\sigma}^{-i}(G)]}[U].$$

⁶Although the *do*-operator is applied in the original paper, it is relaxed here to allow for the implementation of more dynamic strategies.

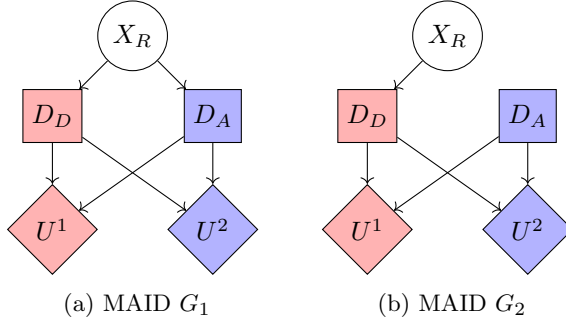


Fig. 8: The different causal structures G_1 and G_2 of the causal Bayesian game as in Example 8.

Fig. 9: The utilities of the deterring agent (left) and attacking agent (right) for causal structure G_1 (top) and G_2 (bottom).

$U^1(G_1)$	$D_A = a$	$D_A = \neg a$	$U^2(G_1)$	$D_A = a$	$D_A = \neg a$
$D_D = d$	-1000	1	$D_D = d$	-1000	-1
$D_D = \neg d$	-1	0	$D_D = \neg d$	1	0

$U^1(G_2)$	$D_A = a$	$D_A = \neg a$	$U^2(G_2)$	$D_A = a$	$D_A = \neg a$
$D_D = d$	-1	1	$D_D = d$	0	-1
$D_D = \neg d$	-1	0	$D_D = \neg d$	1	0

Example 8 (Causal Bayesian Deterring Game) Suppose two agents have their private beliefs about the causal structure of the game they are playing. In both games, players make decisions to defend D_D and attack D_A independently. While the utility of both agents is the result of both agents' actions, the attacking agent's decision can also be shaped by the defending agent's capability to retaliate X_R . Figure 8 illustrates the two different causal structures under consideration. While the attacking agent does not have access to the defending agent's retaliation capacity nor does he think the defending agent thinks he has ($\mu_A(G_2) = 1$ and $\mu_A(\mu_D(G_1)) = 1$), the defending agent considers a scenario where the attacking has access to his retaliation capacity with equal probability: $\mu_D(G_1) = \mu_D(G_2) = \mu_D(\mu_A(G_1)) = \mu_D(\mu_A(G_2)) = \frac{1}{2}$. Taking into account the utilities for the different structures indicated by Table 9, the Bayesian Nash equilibria are $((\neg a, a), d)$ and $((a, a), \neg d)$. These are found by checking all possible strategy combinations and identifying those where: the defender's action is optimal given their 50-50 uncertainty about which type of attacker they face, each attacker type's action is optimal given their own beliefs about the game structure, and no player would want to change their action given what the others are doing.

While this game and associated Bayesian Nash equilibrium is similar to Example 4, we emphasize that uncertainty about the causal structure in this example only gives rise to different pay-offs. The key addition of the causal Bayesian game is that agents reason not only about uncertainty over payoffs,

but about uncertainty over the causal mechanisms that generate those payoffs, enabling analysis of how interventions would affect outcomes differently under each hypothesized structure. When alternative causal structures yield distinct payoff configurations and more sophisticated higher-order beliefs are involved, significant complications may arise. Such complexity, and the associated computational burden, may account for why no implementation for computing Bayesian Nash equilibria in causal games currently exists.

4.4 Practical Guide to Causal Game Theory

Analogous to selecting a game-theoretic model, the choice of a causal game-theoretic model should consider whether agents possess private information regarding the causal structure. As the mathematical dissection discerns different types of nodes within causal games, practitioners must also clearly differentiate between decisions, chance events, and utilities. This distinction can be subtle, as illustrated in Example 7: a deterrer’s capability, often modeled as a chance node, may not truly qualify as such if the agent has the option to enhance their capabilities. Consequently, this classification requires careful consideration, thoughtfully aligned with the specific research question at hand.

Unlike the standard causal graphs in Example 5, which consist solely of chance nodes, causal games with decision nodes do not require the elicitation of conditional probability distributions for those decisions, as they are being solved in response to the adversary. This game-theoretic aspect in causal games

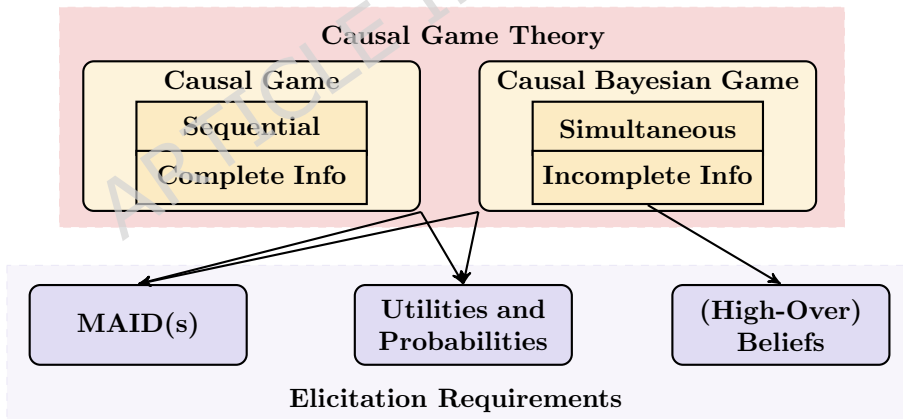


Fig. 10: Elicitation requirements for causal games and causal Bayesian games and associated characteristics: a causal game necessitates the elicitation of a MAID, utilities, and conditional probabilities. In contrast, a causal Bayesian game further requires the elicitation of multiple MAIDs along with beliefs about the probabilities over these graphs and higher-order beliefs about other players’ beliefs.

thus reduces some of the elicitation burden. The remaining conditional distribution of the chance nodes and the specifications in the utility nodes can be extracted via the elicitation methods introduced in Section 3.4 and 2.5, respectively.

While extracting higher-order beliefs in addition to uncertainty over the nature of graphical structure may appear highly complex, the methods for eliciting prior and posterior probabilities outlined in Section 2.5 remain applicable (Mikkola et al, 2023). However, the increased complexity associated with calculating the relevant solution concepts across different causal structures may impede practical implementation.

A summary of the causal game and the causal Bayesian game along with required information for implementation is given in Figure 10.

5 Future Directions

Building on the historical development traced in the introduction, this paper makes three distinct contributions. First, we provide a unified framework that synthesizes these developments within a common probabilistic graphical model formalism, articulating the distinctions between models, their assumptions, and required inputs. Second, by consistently adapting a practical security-domain example and curating relevant implementation research, we bridge the divide between theoretical foundations and applied practice. Third, we equip practitioners with structured guidance for model selection and application, fostering closer collaboration between researchers and practitioners.

We propose the following research directions. First, while tools exist for implementing the introduced models and conducting their respective elicitation separately (James Fox et al, 2021; Barons et al, 2022), we recommend developing an integrated elicitation and modeling tool to further streamline the creation of such applications. Second, although causality has been extended to incorporate temporal dynamics (Runge et al, 2023), no such dynamic causal models, to the best of our knowledge, have been adapted for game-theoretic contexts. Such an extension, though demanding in terms of elicitation requirements, could better capture complex decision-making dynamics.

The introduced frameworks also share deep connections with causal reinforcement learning and multi-agent reinforcement learning, where structural causal models provide a common foundation for reasoning across Pearl’s causal hierarchy (Bareinboim et al, 2024). The most profound intersection resides at the counterfactual layer: credit assignment methods in multi-agent reinforcement learning (Foerster et al, 2018) and equilibrium refinements in causal game theory both fundamentally require reasoning about what would have happened under alternative actions. This shared reliance on counterfactual reasoning suggests a promising avenue for cross-pollination between these fields.

We hope that our research paves the way for more applications that harness the combined strengths of strategic and causal reasoning.

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